ECE264D:
CMOS Analog Integrated Circuits and Systems IV

Lecture 6:
Oscillators

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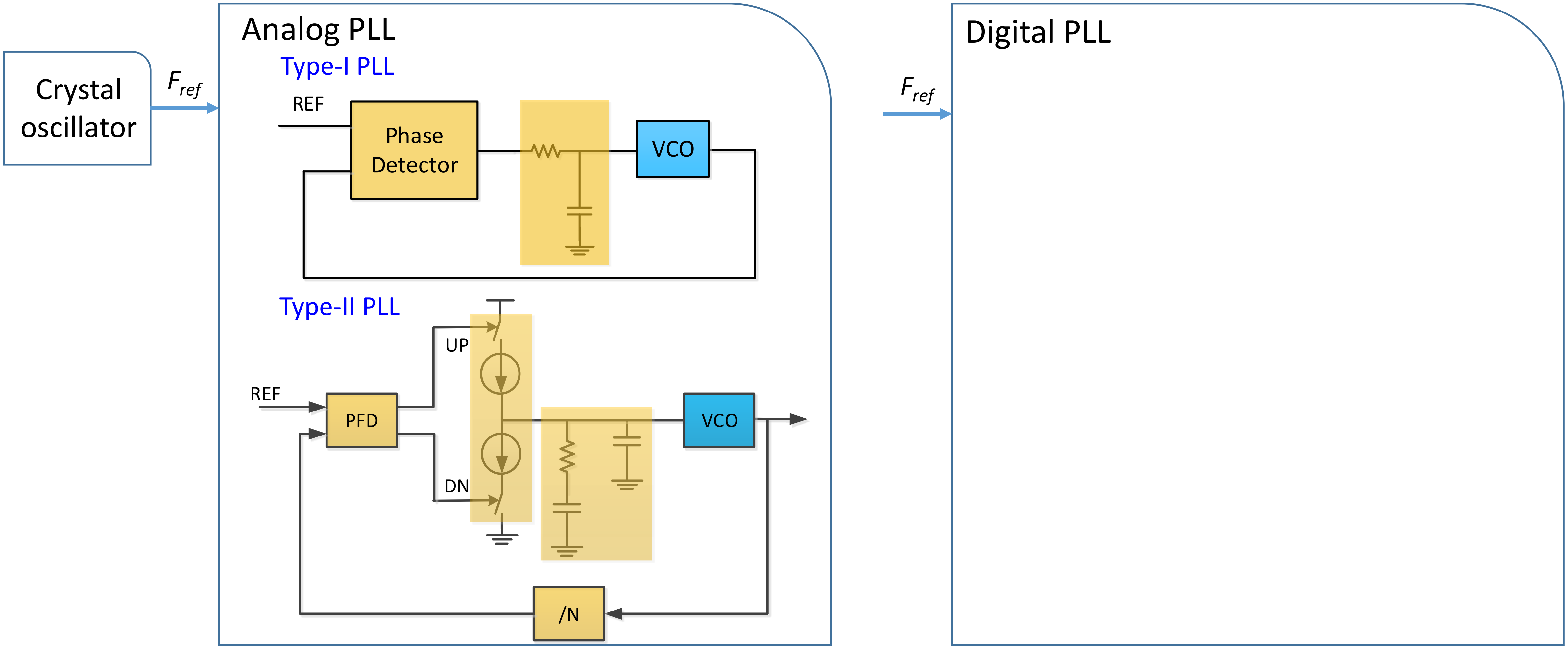
Frequency Synthesizer Overview



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Outline

- VCO specifications
- Oscillator theory of operation
  - Oscillation start up condition
  - Feedback system model
  - One port model
- Oscillator tuning
  - Continuous tuning with varactor
  - Capacitor bank tuning
  - Inductive tuning
  - VCO gain linearization

Oscillator Specifications

- Frequency tuning range
- Phase noise
  - Close-in phase noise
  - In-band integrated phase noise (IPN)
  - Out of band phase noise
  - Far out phase noise
- VCO gain (K_{VCO})
- Power consumption
- Supply sensitivity (K_{Vdd})
- Figure of merit (FOM)

VCO Figure of Merit

$$FOM = 10log\left[\left(\frac{f_0}{\Delta f}\right)^2 \frac{1}{L(\Delta f)P_{DC,mW}}\right]$$

- $L(\Delta f)$ is the phase noise at carrier frequency of f_0 and offset frequency of Δf.
- $P_{DC,mW}$ is the power consumption in mW.
- FOM is used to normalize VCO performance for comparison
- Even after normalization, FOM is worse at low offset frequency (e.g. 100kHz due to flicker noise
- FOM is worse at high oscillation frequency (e.g. millimeter wave VCO) due to tank Q degradation at high frequency.

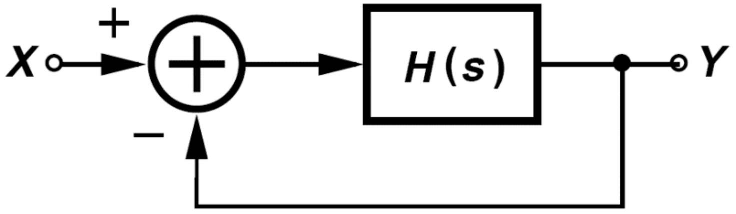
	This Work		ISSCC'17 [C-C. Li et al.]	JSSC'17 [2]	ISSCC'15 [4]	JSSC'13 [M. Babaie et al.]	JSSC'08 [A. Mazzanti et al.]	JSSC'01 [1]
Topology	**CMOS Inverse-Class-F**		Trifilar Coil	NMOS CM resonance	Class-$F_{2,3}$	Class-F	Class-C	NMOS+2F_{LO} tail inductor
CMOS Tech.	65nm		16nm	28nm	40nm	65nm	130nm	0.35μm
Tuning Range	3.49-4.51 (25.5%)		3.2-4.0 (22%)	2.85-3.75 (27.2)	5.4-7 (25%)	2.95-3.8 & (25%)	4.9-5.65 (14%)	1-1.2 (18%)
V_{DD} (V)	0.6		0.4	0.9	1	1.25	1	2.5
Freq. (GHz)	3.49	4.51	3.23	2.89	5.4	3.7	4.9	1.2
Power (mW)	1.2	1.14	3.8	6.8	12	15	1.4	9.1
PN (dBc/Hz) @ 100k/10MHz	-102.4/ -145.6	-98.5/ -143.7	-99 */ -145	-108.1/ -152	-105.3/ -146.7	-103.6/ -152.8	-99 */ -142 #	-117 */ -162.5 #
FOM (dBc/Hz) @ 100k/10MHz	**192.5/ 195.6**	**191/ 196.2**	183.3/ 190	188/ 192.5	189.1/ 190.5	183.2/ 192.4	191.3/ 194.4	189/ 195
Freq. Pushing (MHz/V)	**4.5**	**15**	3.6-27.3	6-30	12-23	18-50	N/A	N/A
$1/f^3$ Corner (kHz)	100	300	120	200	60	700	200	N/A
Die Area (mm²)	0.14		0.11	0.19	0.13	0.12	N/A	N/A

* Estimated from PN plot # Normalized from 1 MHz offset & After on-chip div-by-2

$FOM = -PN + 20\log_{10}(f_0/\Delta f) - 10\log_{10}(P_{DC}/1mW)$

Ref[5]: C-C Lim, *et al.*, ISSCC 2018, 23.5, p374-376.

Oscillator Startup Condition



Feedback system model

$$\frac{Y}{X}(s) = \frac{H(s)}{1 + H(s)}$$

- Barkhausen's criteria for oscillation

$$|H(s = j\omega_1)| = 1, \angle H(s = j\omega_1) = 180^o$$

- At frequency ω_1, signal going through H(s) has a gain of 1 and a phase shift of 180^o.
- Open loop transfer function has a gain margin or phase margin of zero.
- Together with the "-" sign that feeds back *Y*(s) to the input, the loop becomes positive feedback.
- With infinitesimal input *X* (such as noise), the loop will amplify it and the output *Y* will grow.

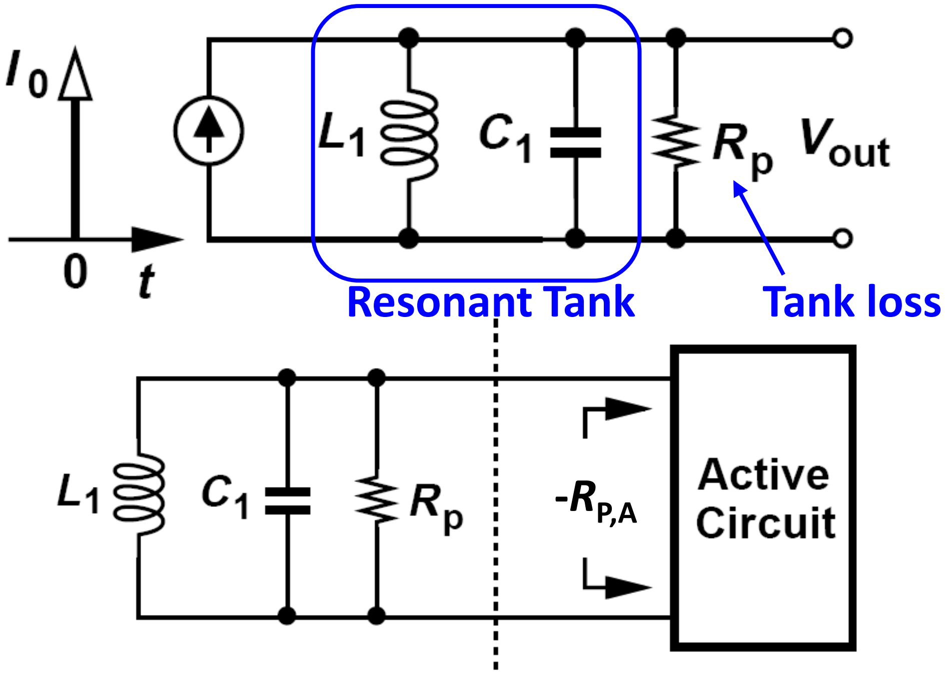
- Two conditions for oscillation

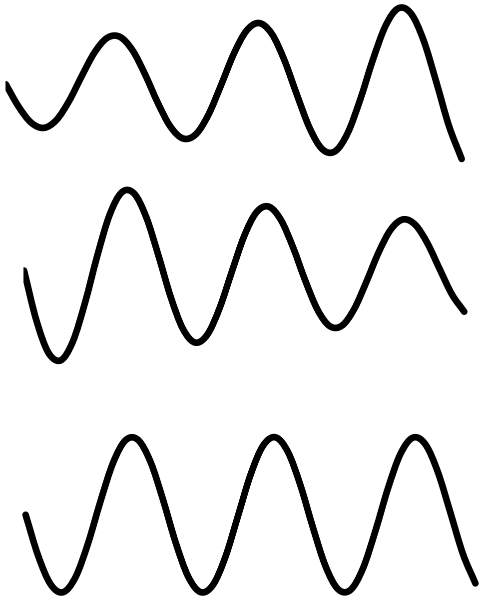
1) $|H(s = j\omega_1)| = 1$ ⇨ Constant amplitude, oscillation self-sustaining

In the actual design, $|H(s = j\omega_1)| > 1$ to have reliable startup and amplitude growth. An **active circuit** is used to generate the gain, start and maintain oscillation

2) $\angle H(s = j\omega_1) = 180^o$ ⇨ Set oscillation frequency by a **passive resonant circuit**, at which exact 180^o phase shift is generated to obtain 360^o phase shift around the loop.

One-Port Model of Oscillator





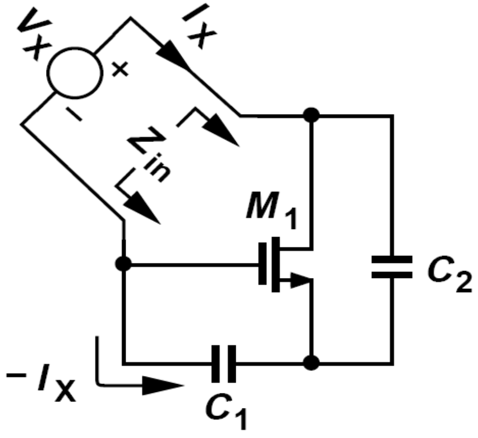
- Active circuit generates $|-R_{P,A}| < R_P$
- Oscillation amplitude grows

- $|-R_{P,A}| > R_P$
- Oscillation amplitude declines

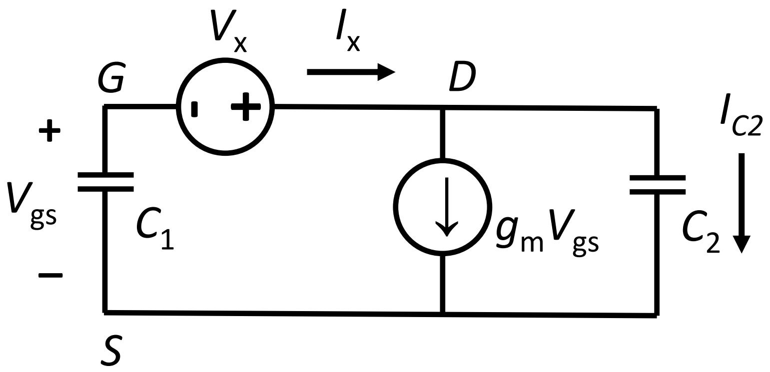
- $|-R_{P,A}| = R_P$
- Oscillation amplitude maintains

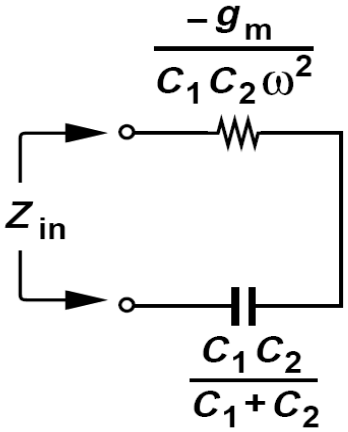
- Passive circuit L1 and C1 form a resonant tank and determine the oscillation frequency: $\omega_1 = \frac{1}{\sqrt{L_1 C_1}}$

One-Port Oscillator



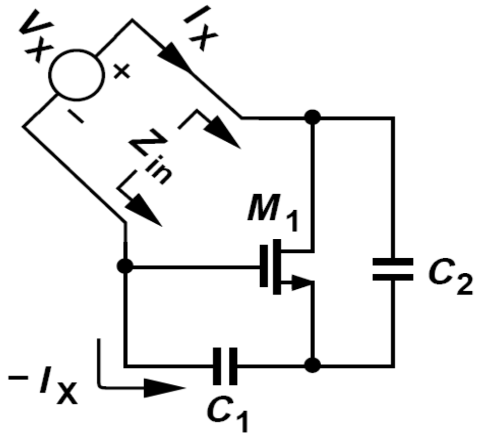
Colpitts Oscillator

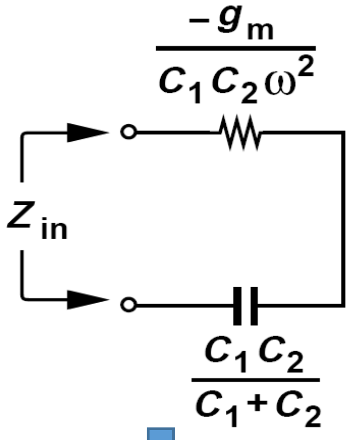




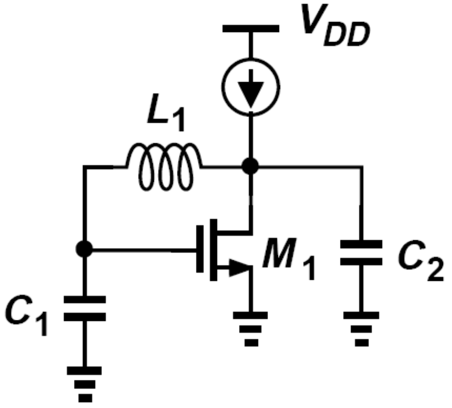
1. $V_{gs} = -\frac{1}{sC_1} I_X$
2. $I_{C_2} = (V_X + V_{gs})sC_2 = \left(V_X - \frac{1}{sC_1} I_X\right) sC_2$
3. $I_X = g_m V_{gs} + I_{C_2} = -g_m \frac{1}{sC_1} I_X + \left(V_X - \frac{1}{sC_1} I_X\right) sC_2$
4. $I_X\left(1 + g_m \frac{1}{sC_1} + \frac{sC_2}{sC_1}\right) = V_X sC_2$
5. $Z_{in} = \frac{V_X}{I_X} = \frac{1}{sC_1} + \frac{1}{sC_2} + \frac{g_m}{s^2 C_1 C_2} = \frac{1}{sC_1} + \frac{1}{sC_2} - \frac{g_m}{\omega^2 C_1 C_2}$

One-Port Oscillator

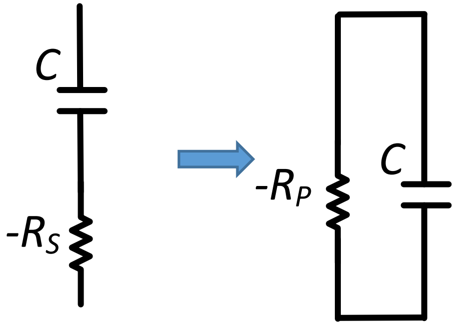




Adding L_1 to form resonant tank
Adding bias current for M_1



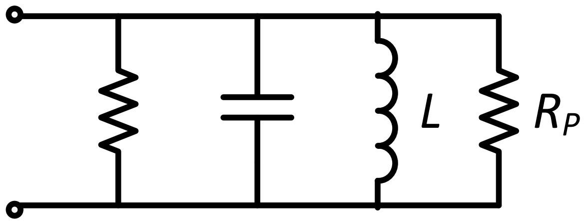
Colpitts Oscillator



$$C = \frac{C_1 C_2}{C_1 + C_2}$$

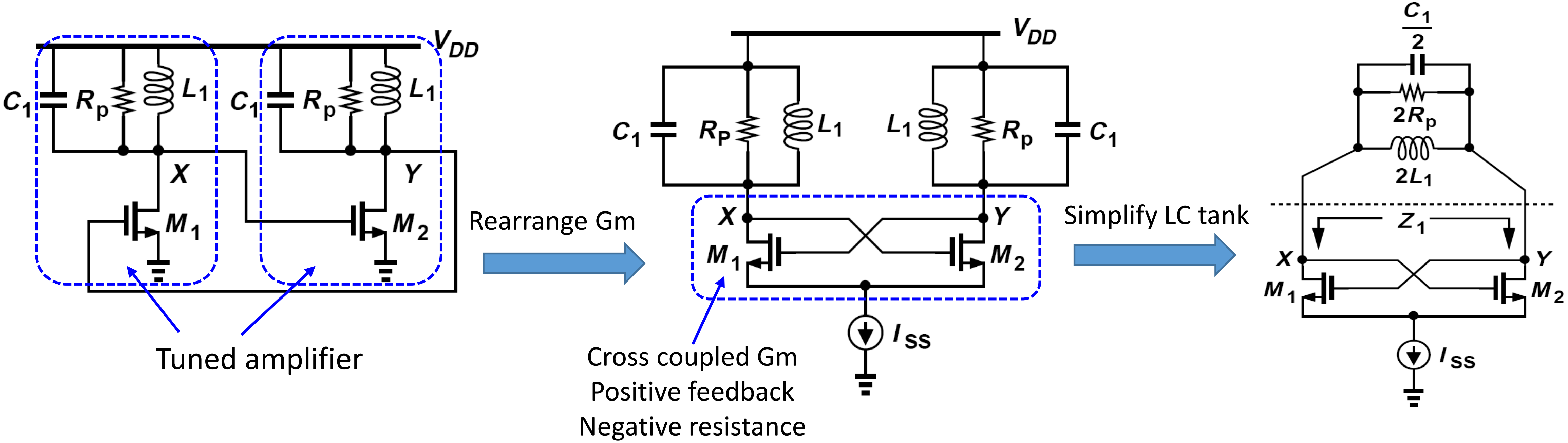
$$-R_P = -\frac{(C_1 + C_2)^2}{C_1 C_2 g_m}$$

- $Z_{in} = \frac{V_X}{I_X}(j\omega) = \underbrace{\frac{1}{j\omega C_1} + \frac{1}{j\omega C_2}}_{C_1 \text{ and } C_2 \text{ in series}} - \underbrace{\frac{g_m}{C_1 C_2 \omega^2}}_{\text{Negative resistance } R_S}$
- Oscillation conditions
  - $\frac{(C_1+C_2)^2}{C_1 C_2 g_m} \leq R_P$
  - $\omega_{osc} = \frac{1}{\sqrt{L\frac{C_1 C_2}{C_1+C_2}}}$



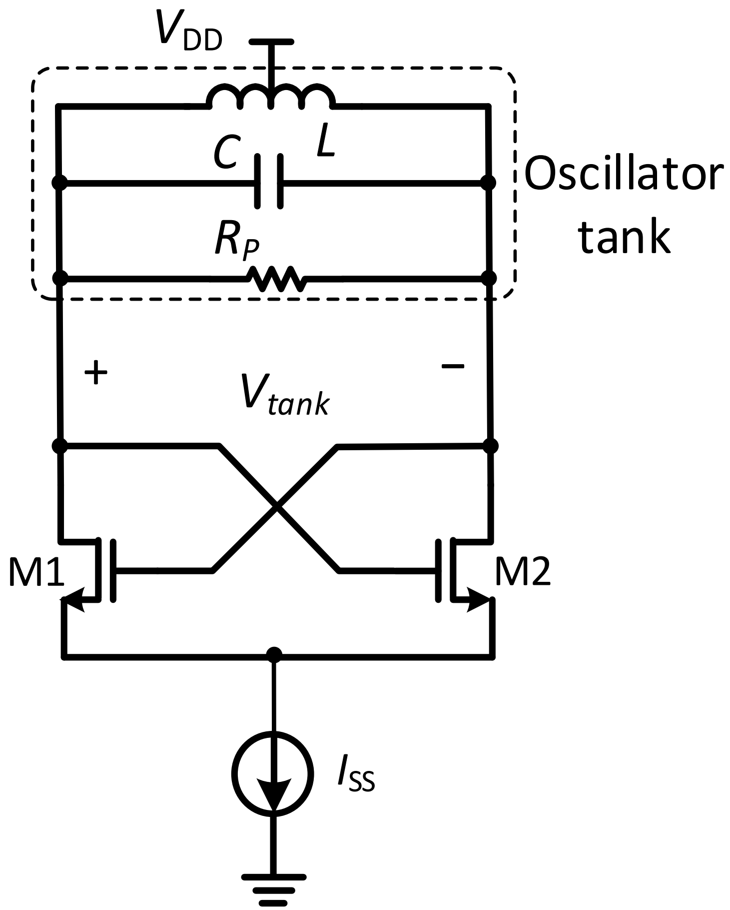
$-\frac{(C_1 + C_2)^2}{C_1 C_2 g_m}$ $\frac{C_1 C_2}{C_1 + C_2}$

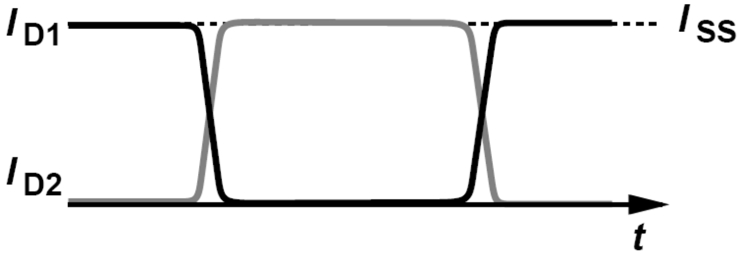
Cross Coupled LC Oscillator



- Negative resistance: $Z_1 = -\frac{2}{g_m}$
- Oscillation startup condition $\frac{2}{g_m} \leq 2R_P \Rightarrow g_m R_P \geq 1$
- Oscillation frequency $\omega_{osc} = \frac{1}{\sqrt{L_1(C_1+C_{para})}}$

Cross Coupled LC Oscillator





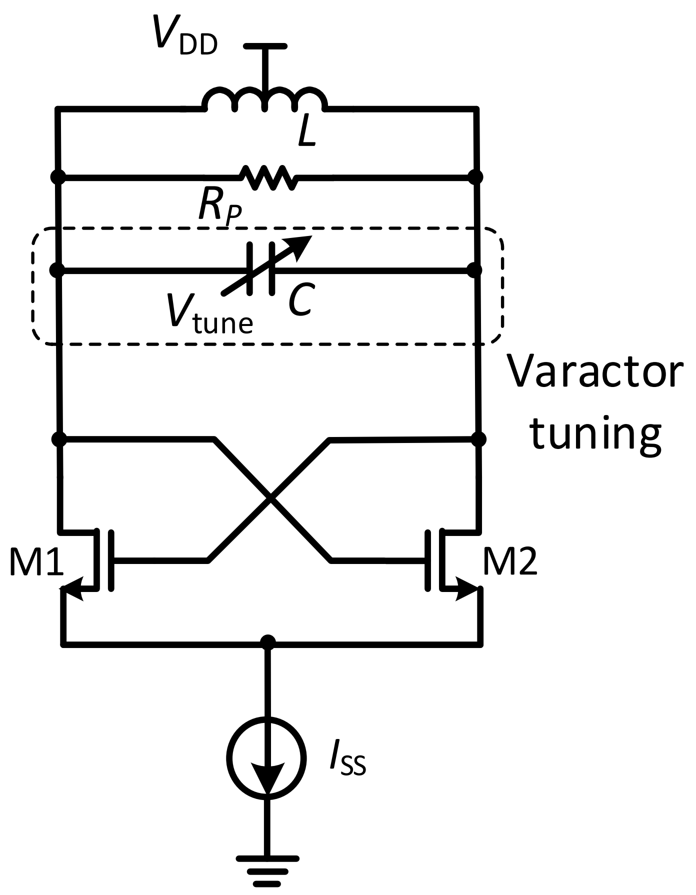
Steady state oscillation amplitude

- Current limited regime
  - M_1 and M_2 completely switch on and off.
  - Tail bias current steers between M_1 and M_2 nearly as square wave.
  - I_{D1} and I_{D2} each has an average of $I_{SS}/2$ and an amplitude of $I_{SS}/2$.
  - First harmonic of the current is multiplied by tank impedance R_P and higher harmonics are attenuated.
  - Tank oscillation amplitude (Differential, 0-to-peak)

$$V_{tank} = \frac{2}{\pi} I_{SS} R_P$$

- Voltage limited regime: $V_{tank} = V_{DD} - V_{dsat}$

Frequency Tuning – Continuous Tuning with Varactor



$$\omega_{osc} = \frac{1}{\sqrt{LC}}$$

VCO gain:

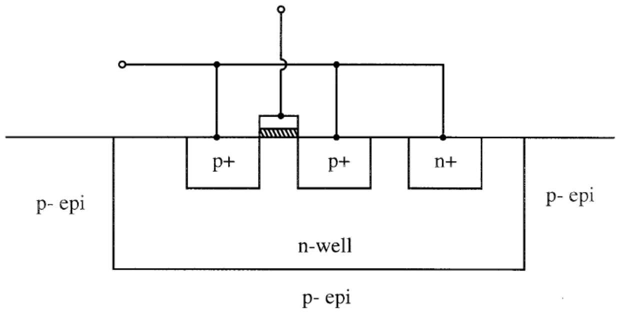
$$K_{VCO} = \frac{d\omega_{osc}}{dV} = \frac{d\omega_{osc}}{dC} \cdot \frac{dC}{dV}$$

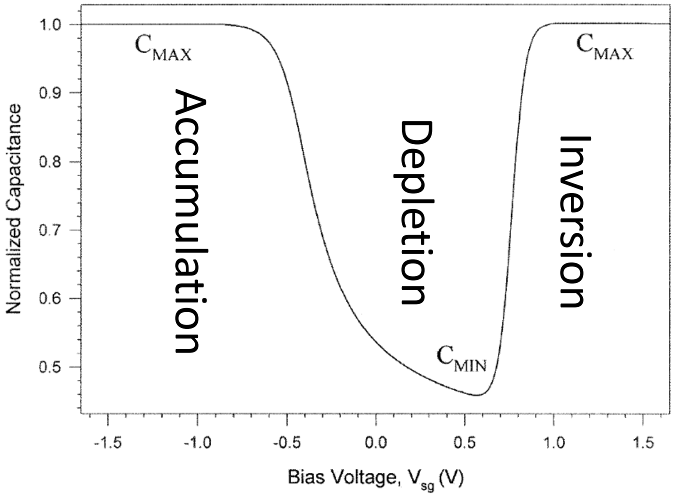
$$= -\frac{1}{2} L \frac{1}{\left(\sqrt{LC}\right)^3} \frac{dC}{dV} = -\frac{L}{2} \cdot \omega_{osc}^3 \frac{dC}{dV}$$

- Varactor design considerations
  - Cover sufficient tuning range (F_{min}, F_{max})
  - Maximize varactor Q
  - K_{VCO} variation: impacts on loop bandwidth, loop dynamics, VCO phase noise, and phase modulation
    - K_{VCO} vs. Vtune (tuning linearity)
    - K_{VCO} vs. oscillation frequency
  - K_{VCO} polarity

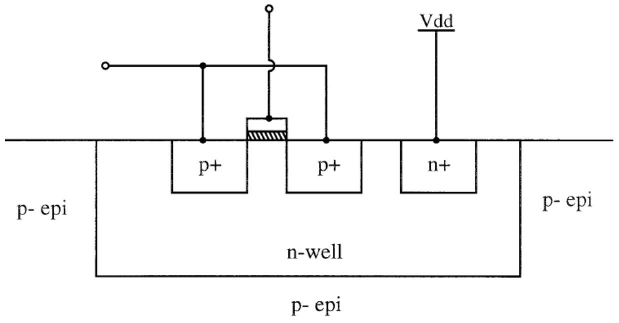
Varactor Types and Characteristics

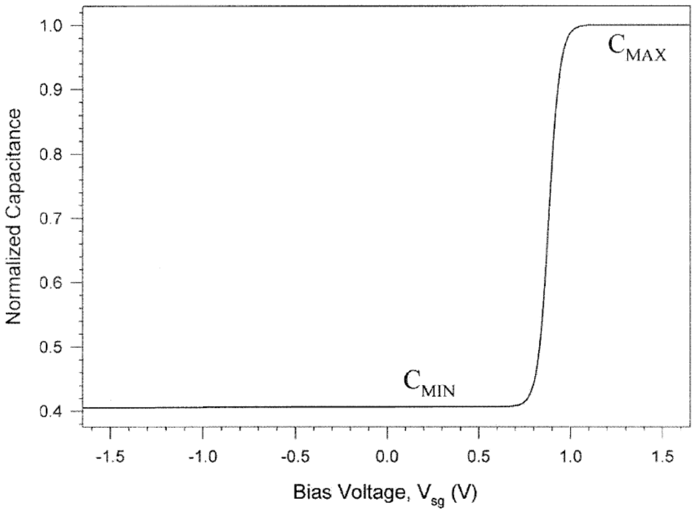
Ref [4]





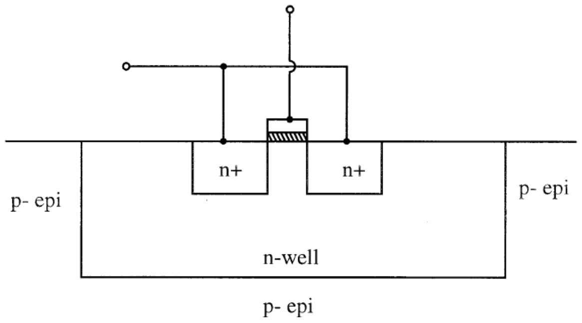
- As V_{sg} ↑, the capacitor goes from accumulation region to depletion and inversion region.
- $C_{max} \approx C_{ox}$
- $C_{min} \approx C_{ox} // C_{pn}$
- C ~ V not monotonic

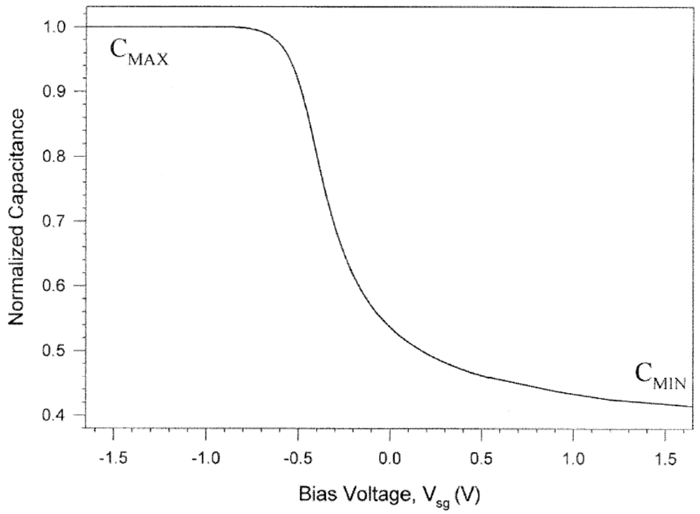




- N-well is connected to Vdd, so only inversion region and depletion region.
- $C_{max} \approx C_{ox}$
- $C_{min} \approx C_{ox} // C_{pn}$
- C ~ V monotonic

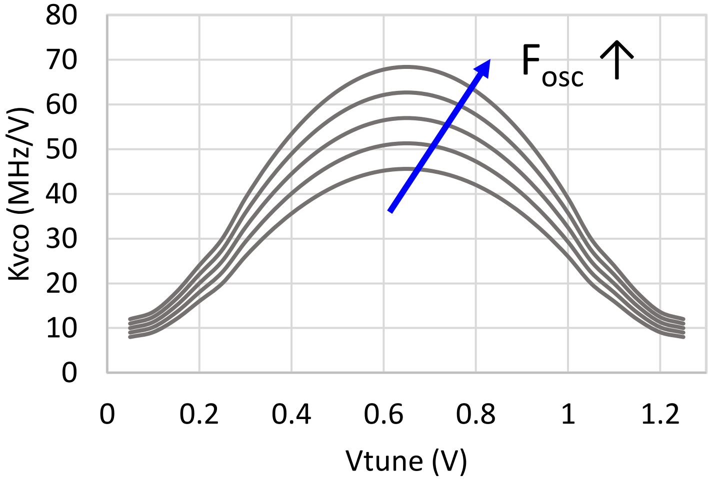
Varactor Types and Characteristics





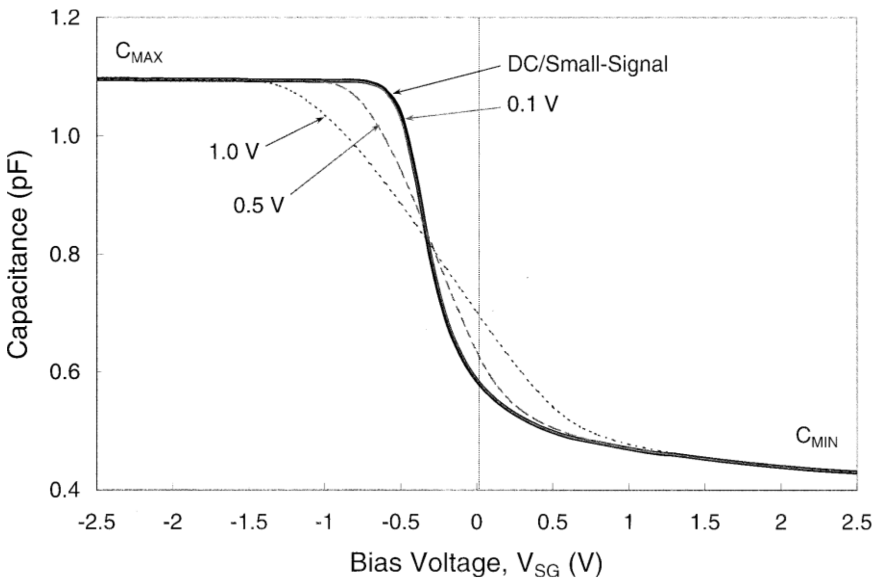
- NMOS in N-well structure
- S/D as well contact. No dedicated well contact
- Only accumulation region and depletion region.
- $C_{max} \approx C_{ox}$
- $C_{min} \approx C_{ox} // C_{pn}$
- C ~ V monotonic

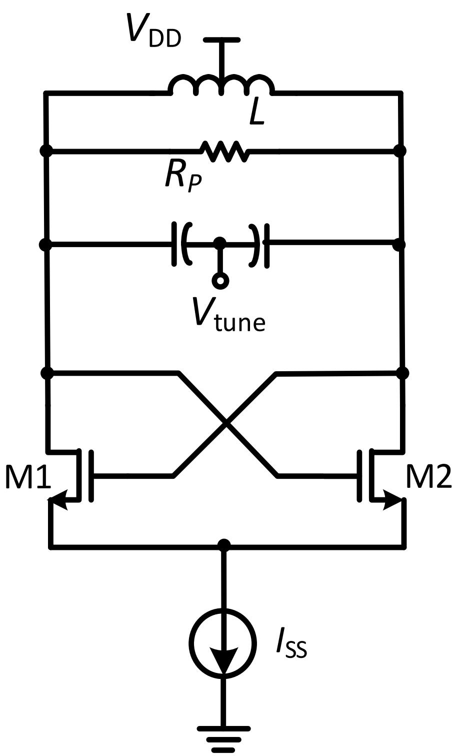
Kvco vs. Vtune at different Fosc

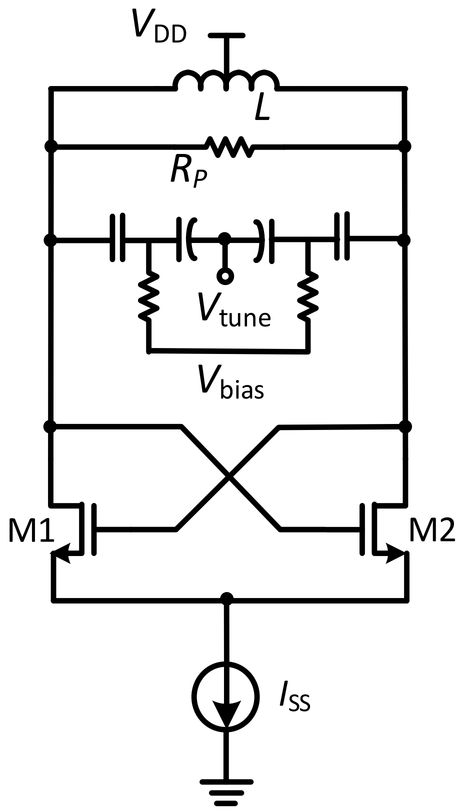


- Channel doping to shift peak C–V slope and Kvco to ~$0.5V_{DD}$
- K_{VCO} is very nonlinear over V_{tune}
- K_{VCO} increases as F_{osc} increases

Varactor Tuning in LC VCO







- Varactor C–V curve represents small signal characteristic
- VCO swing large signal across varactor: C–V curve linearization
  - $i_C = C\big(v(t)\big)\frac{dv(t)}{dt}$, $C_{avg} = \frac{\mathrm{rms}(i_C)}{\mathrm{rms}\left(\frac{dv(t)}{dt}\right)}$
- AC coupled varactor allows a separate V_{bias} independent from V_{DD}
- Noise in varactor bias voltage is critical to VCO phase noise performance

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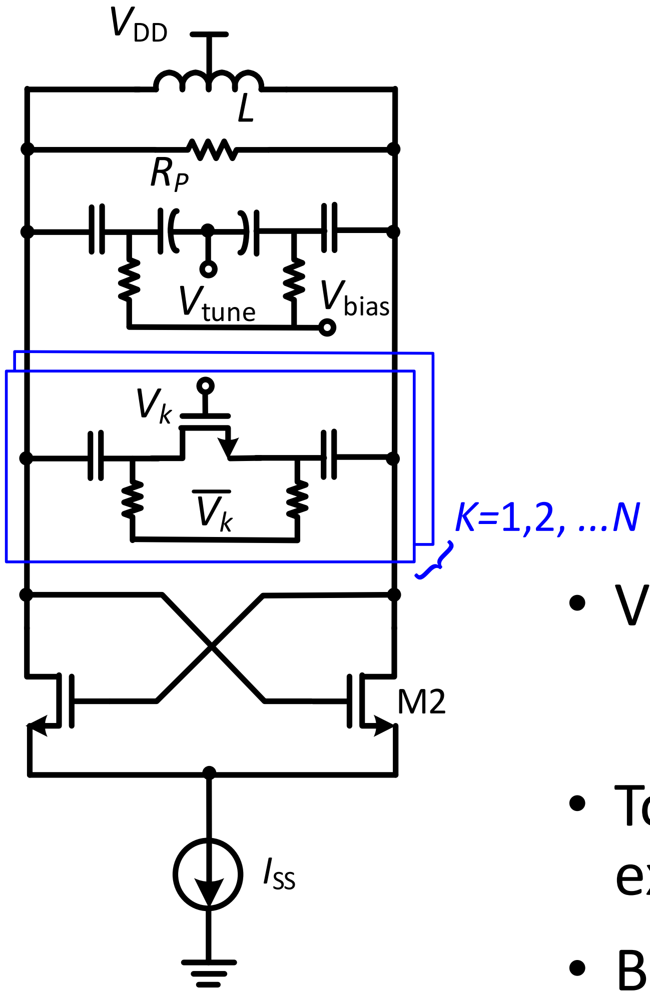
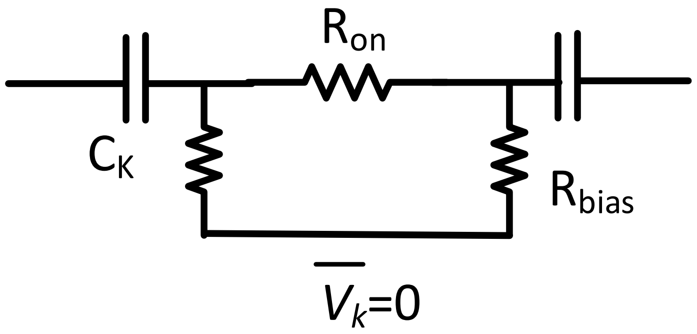
Coarse Tuning with Capacitor Bank

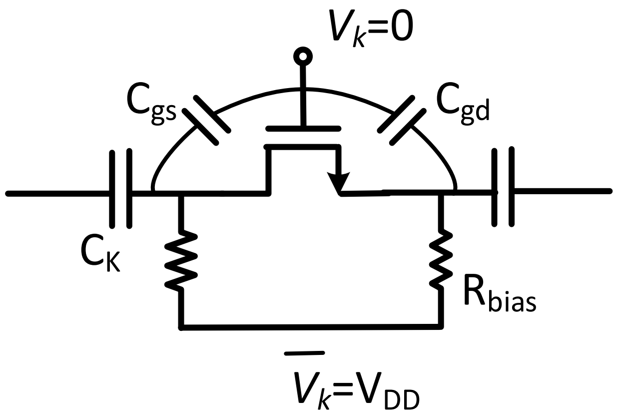


Ref [7]

$V_k=V_{DD}$, $V_{GS}=V_{GD}=V_{DD}$,
$C_{on}=C_K/2$, $R_{on}<<R_{bias}$

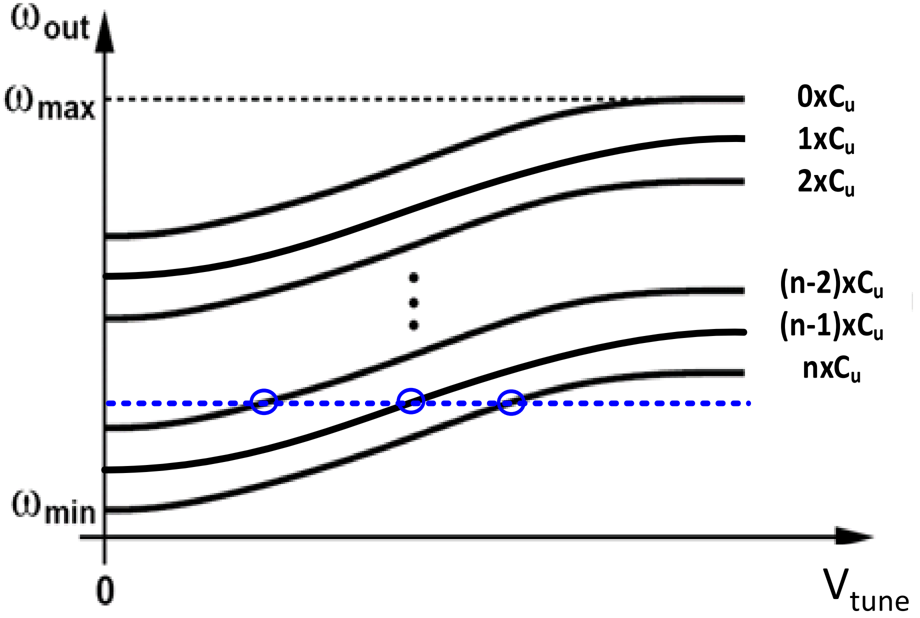


$V_k=0$, $V_{GS}=V_{GD}=-V_{DD}$,
$C_{off}=C_{gs}//C_{gd}//(C_K/2)$



- VCO tuning range
  - $TR(\%) = \frac{f_{max}-f_{min}}{(f_{max}+f_{min})/2} \times 100\%$
- To cover typical tuning range of 30~50%, varactor value and size is excessively large: de-Q the tank and generate AM-PM noise
- Binary weighted or thermometer coded capacitor bank
- Capacitor bank MOS switch S/D reverse biased to reduce parasitic C_{off}

Coarse Tuning with Capacitor Bank



- Capacitor bank for coarse tune and varactor for fine tune
- Binary weighted or thermometer coded capacitor bank
- Any given VCO frequency covered by at least 3 coarse tune curves
  - Reduce V_{tune} variation after lock -> Reduce K_{VCO} variation

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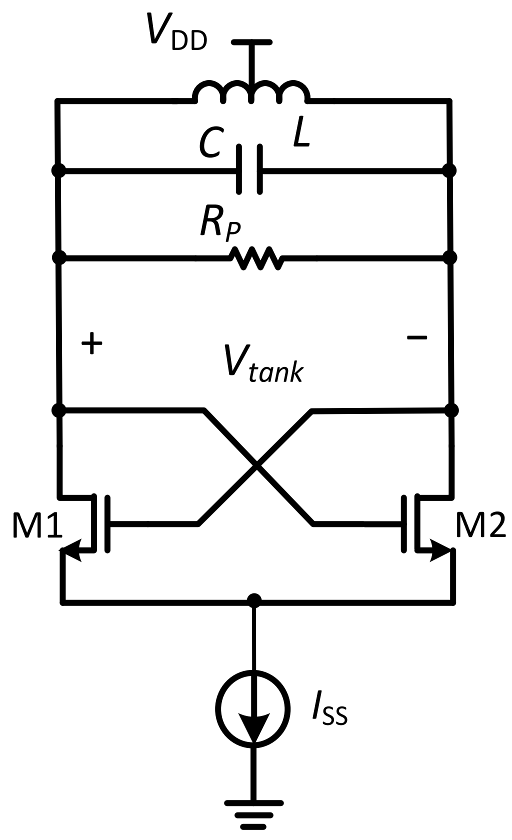
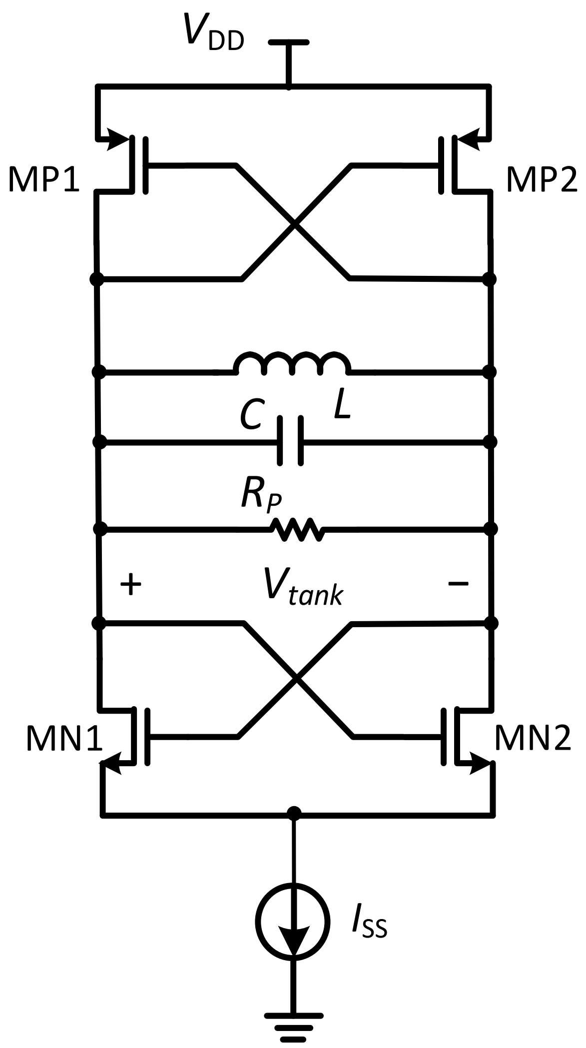
NMOS VCO and CMOS VCO



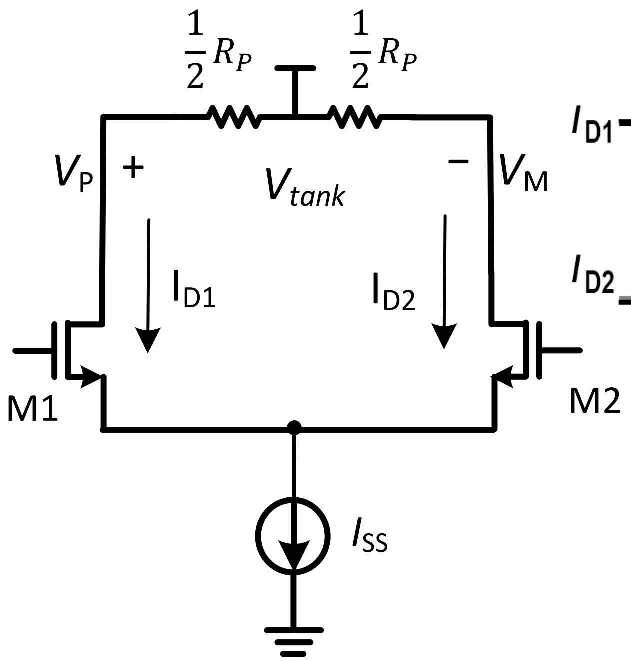
NMOS LC VCO

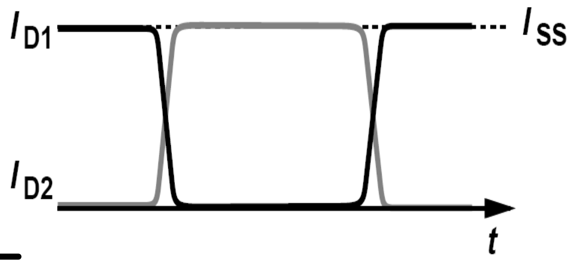
	NMOS VCO	CMOS VCO
Gm devices	NMOS pair	NMOS & PMOS pairs
Swing at same current	$\frac{2}{\pi} I_{SS} R_P$	$\frac{4}{\pi} I_{SS} R_P$
Maximum achievable swing	V_{DD}	$0.5 \cdot V_{DD}$
Parasitic capacitance from Gm	Low	High
PN at same current	High	Low
Best achievable PN at same VDD	Low	High
Startup gain at same current	$g_{mN} R_P$	$(g_{mN} + g_{mP}) R_P$

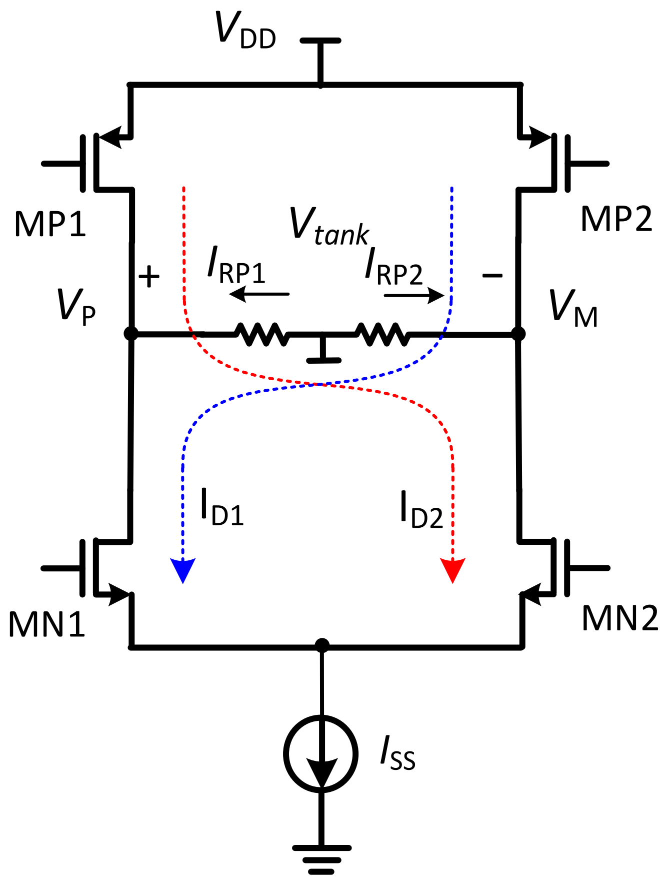


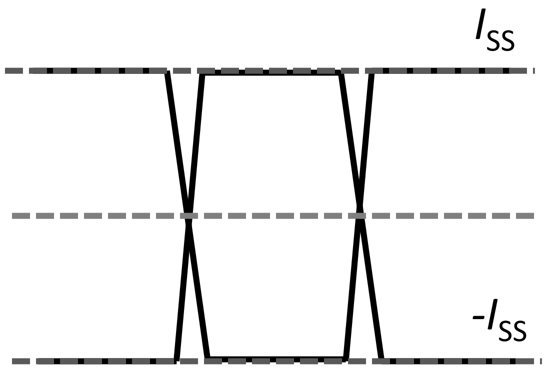
CMOS LC VCO

NMOS VCO and CMOS VCO





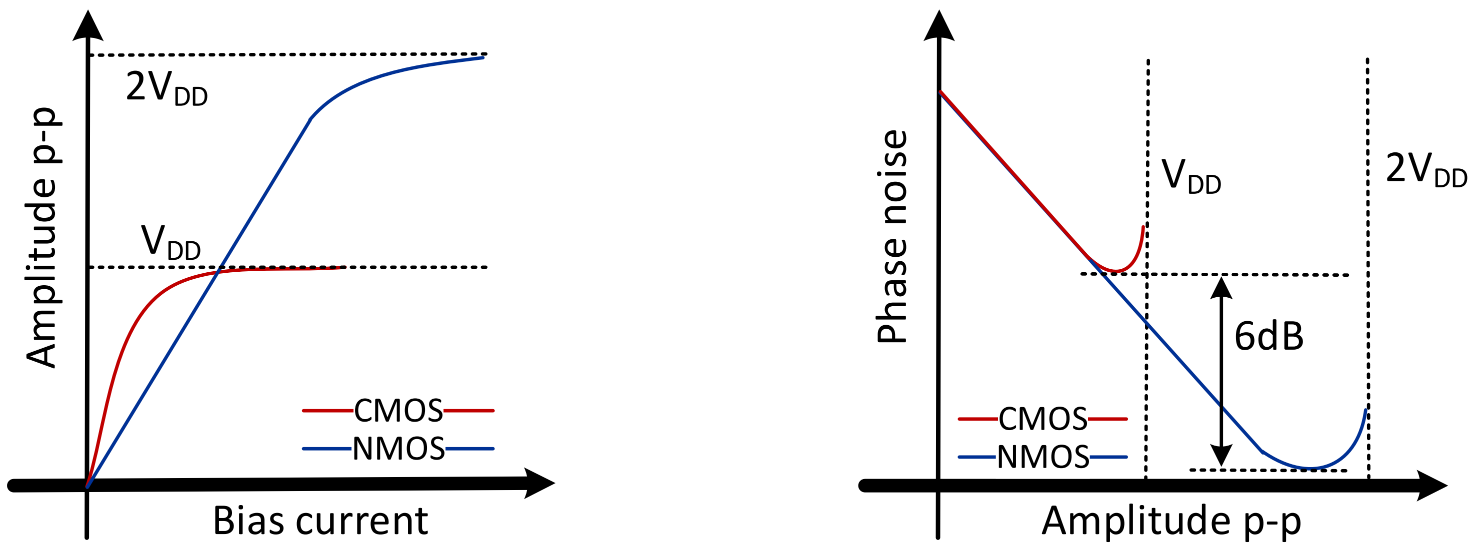




- Center of R_P is virtual AC ground
- Each half R_P experiences nearly square wave current 0~I_{SS}
- Fundamental harmonic of I_{D1} and I_{D2} is $\frac{2}{\pi}I_{SS}$
- V_P or V_M swing is $\frac{2}{\pi}I_{SS}\cdot\frac{1}{2}R_P$
- Differential tank swing (0~peak) is $\frac{2}{\pi}I_{SS}\cdot R_P$

- Center of R_P is virtual AC ground
- Each half R_P experiences nearly square wave current $-I_{SS}\sim I_{SS}$
- Fundamental harmonic of I_{RP1} and I_{RP2} is $\frac{4}{\pi}I_{SS}$
- V_P or V_M swing is $\frac{4}{\pi}I_{SS}\cdot\frac{1}{2}R_P$
- Differential tank swing (0~peak) is $\frac{4}{\pi}I_{SS}\cdot R_P$

NMOS VCO and CMOS VCO



- The choice of NMOS and CMOS VCO depends on multiple factors
  - Phase noise requirement and available VDD level
    - For given VDD, choose CMOS topology if it can meet the phase noise requirement at maximum amplitude (VDD). It requires less current and yields better efficiency and FOM
  - Maximum oscillation frequency: NMOS VCO has less parasitic capacitance and suits better for high frequency VCO, such as millimeter wave VCO.

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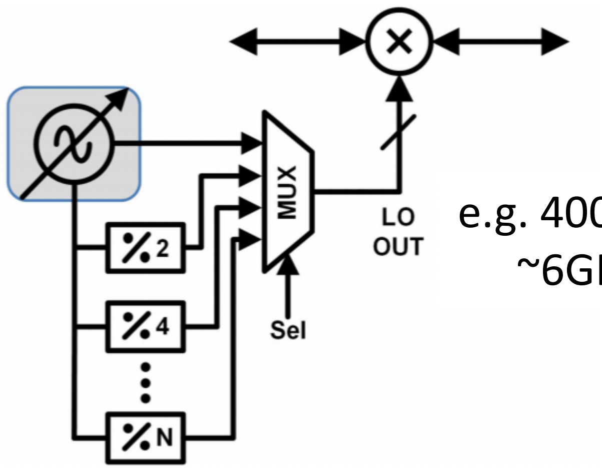
Wide Tuning VCO

Tuning schemes	Tuning range	Purposes	Notes
Varactor tuning	A few tens of MHz (<1%)	PLL loop locking	Continuous tuning; Low Q at mmW frequency
Coarse tune capacitor bank tuning	1~2 GHz (20~40%); Step <10MHz	Full frequency coverage Sufficient overlap	Discrete tuning; Switch size tradeoff between R_{on} and C_{off}
Switched inductor/ magnetic tuning	Multi-GHz (>50%) Step ~GHz	Switch between different band groups; octave tuning	Switch in series with inductor degrades Q
Switched mode tuning	Multi-GHz (>50%) Step ~GHz	Switch between different band groups; octave tuning	Dual core; Reliably enhance one mode and suppress the other

$$\frac{f_{max}}{f_{min}} > 2$$

e.g. 6~12GHz



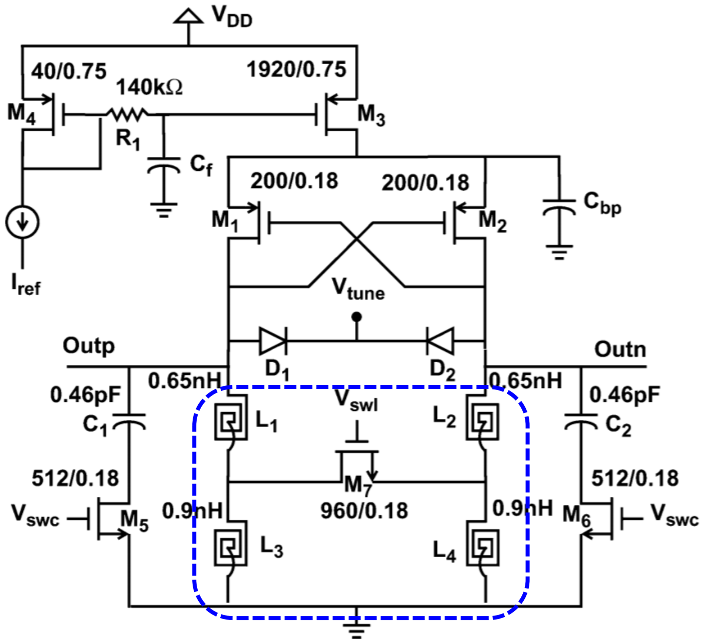
e.g. 400MHz ~6GHz

Ref [6]

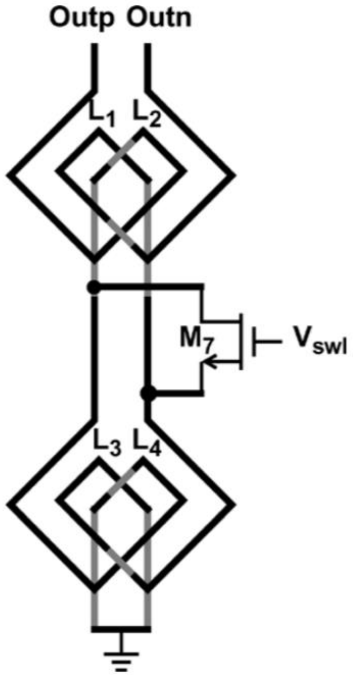
Motivation of wide tuning VCO:

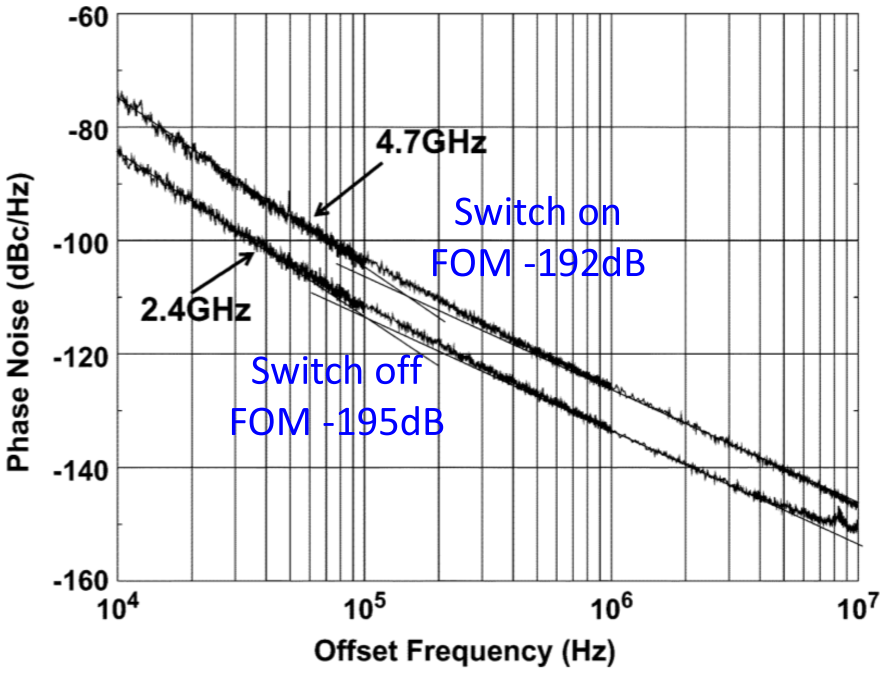
- With an octave VCO, any LO frequency can be covered by different frequency divide ratios
- Reduce number of VCOs needed, saving area

Wide Tuning VCO



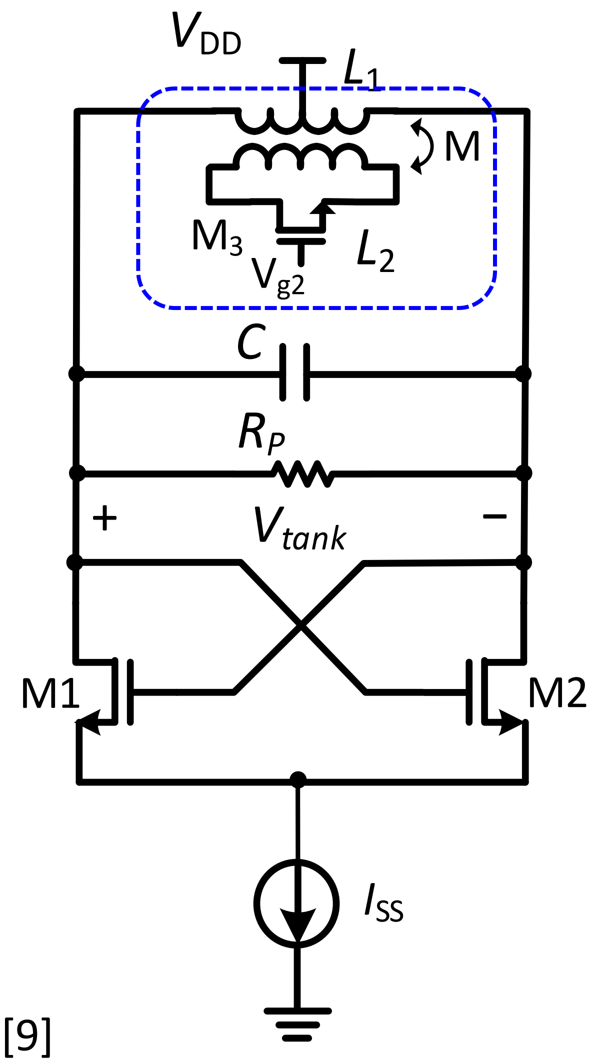
Ref [8]



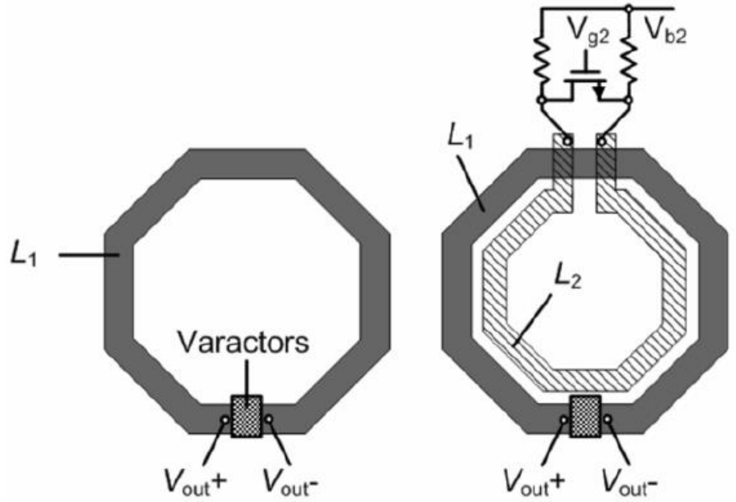


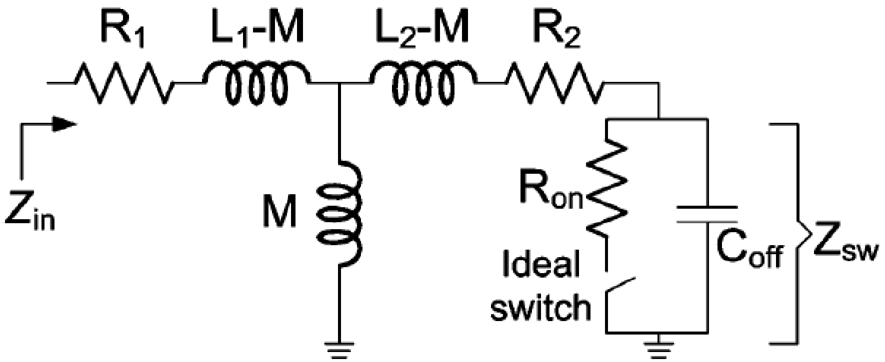
- **Direct inductor switching**
  - Switch (M7) on: tank inductor ≈ L1+L2
  - Switch off: tank inductor = L1+L2+L3+L4
- Switch resistance degrades inductor Q

Wide Tuning VCO



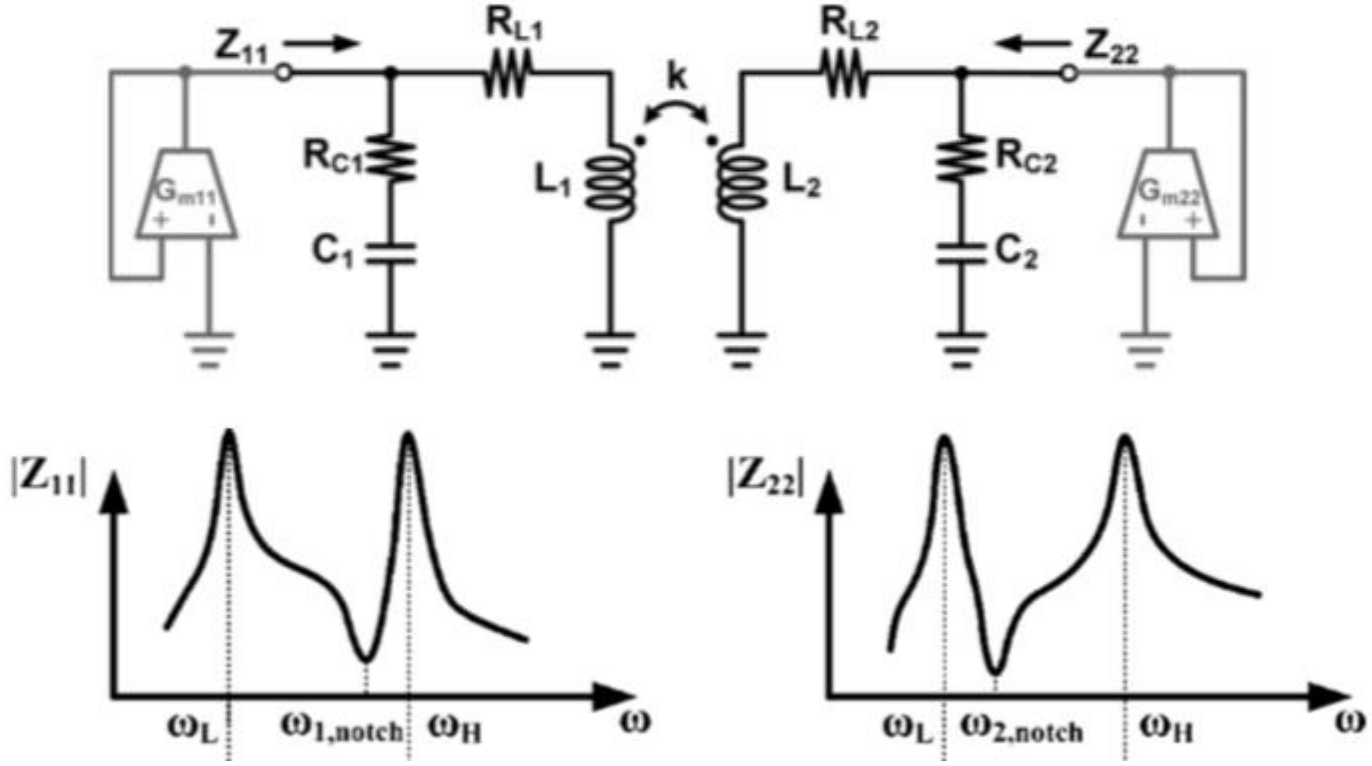
Ref [9]





- **Magnetic switching**
  - Switch (M3) on: tank inductor $L_{eff} \approx L_1 - \frac{M^2}{L_2}$
  - Switch off: tank inductor $L_{eff} \approx L_1$
  - Switch resistance still degrades inductor Q

Wide Tuning VCO



$$Z_{11} \approx \frac{j\omega_1^2 L_1 \omega\left[\left(1-k^2\right)\omega^2 - \omega_2^2\right]}{(k^2-1)\omega^4 + \left(\omega_1^2+\omega_2^2\right)\omega^2 - \omega_1^2\omega_2^2}$$

$$Z_{22} \approx \frac{j\omega_2^2 L_2 \omega\left[\left(1-k^2\right)\omega^2 - \omega_1^2\right]}{(k^2-1)\omega^4 + \left(\omega_1^2+\omega_2^2\right)\omega^2 - \omega_1^2\omega_2^2}$$

Z_{11} and Z_{22} have two peak frequencies:

$$\omega_{H/L}^2 = \frac{\omega_1^2 + \omega_2^2 \pm \sqrt{\left(\omega_1^2 - \omega_2^2\right)^2 + 4k^2\omega_1^2\omega_2^2}}{2(1-k^2)}$$

$$\omega_1 = \frac{1}{\sqrt{L_1 C_1}}, \omega_2 = \frac{1}{\sqrt{L_2 C_2}}$$

- 4^{th} order resonant tank
- Two peak frequencies
- Two modes of oscillation, ω_H and ω_L.

Ref [10]

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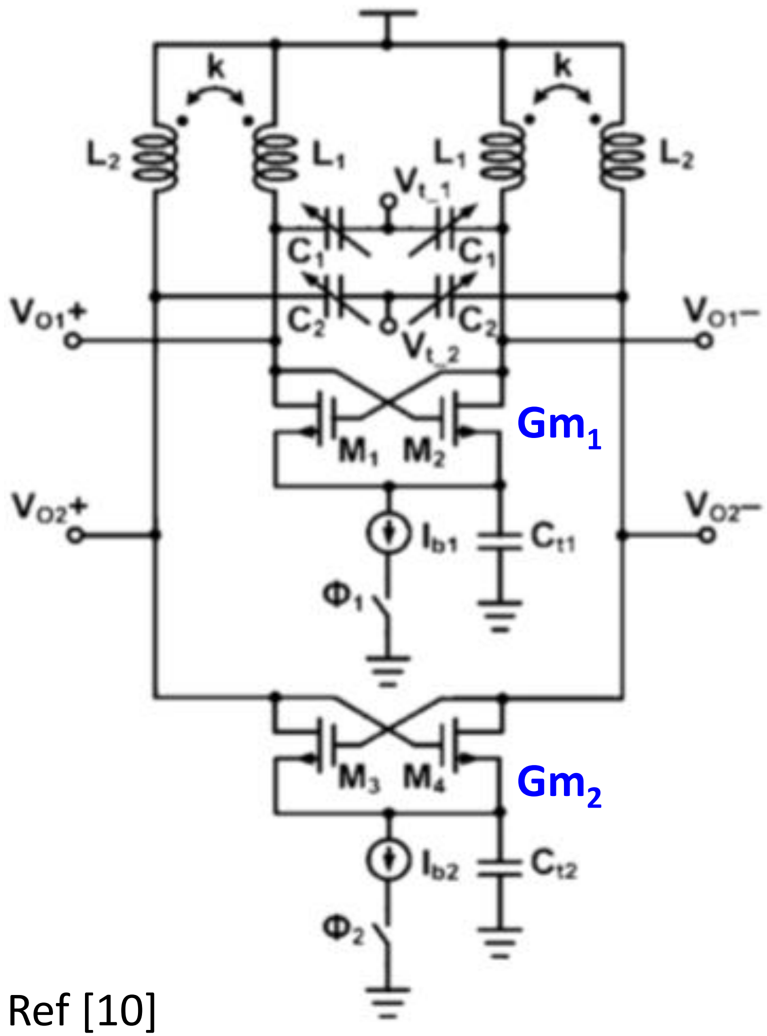
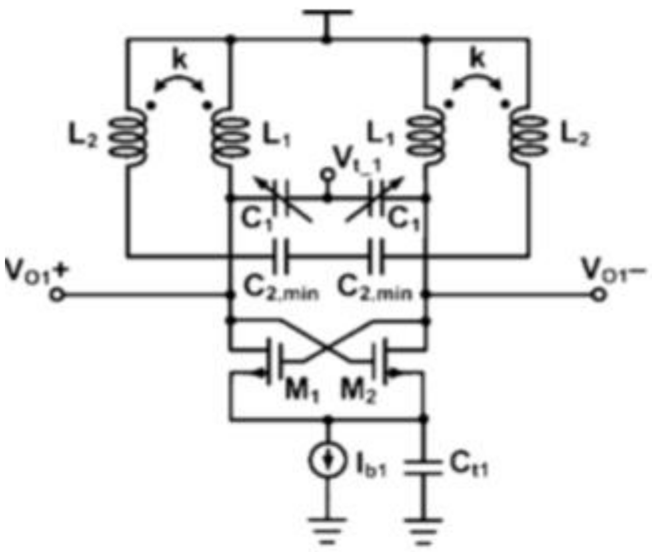
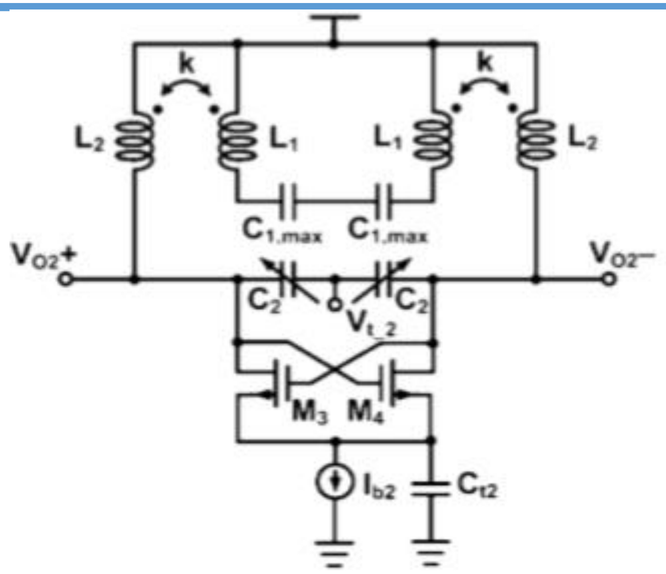
Wide Tuning VCO



Ref [10]



Low band mode, ω_L



High band mode, ω_H

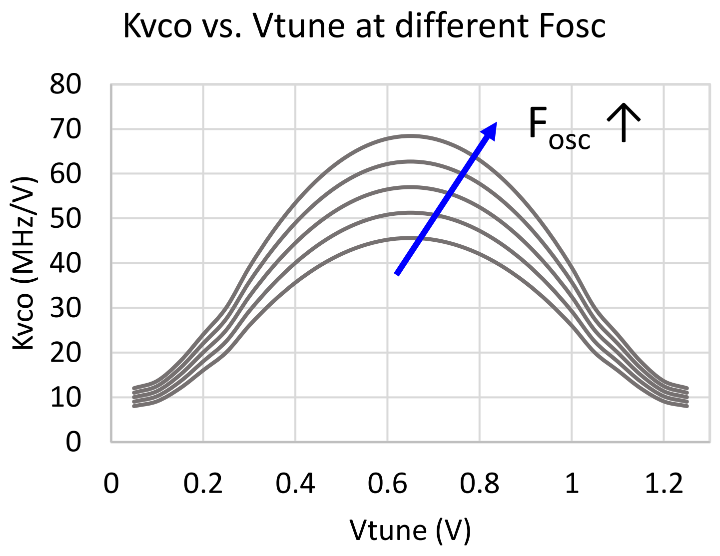
- The oscillation at either ω_L or ω_H can be selected by controlling the bias current sources.
- In low band mode, I_{b1} is enabled while is I_{b2} disabled
- In high band mode, I_{b2} is enabled while is I_{b1} disabled
- Need large ω_1/ω_2 ratio for stable mode selection

VCO Gain Linearization

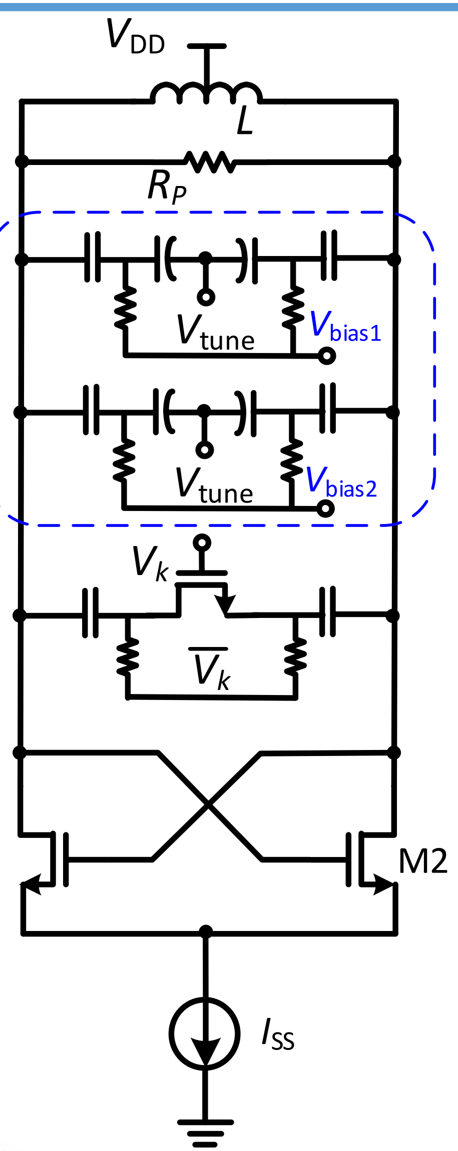
- VCO gain:

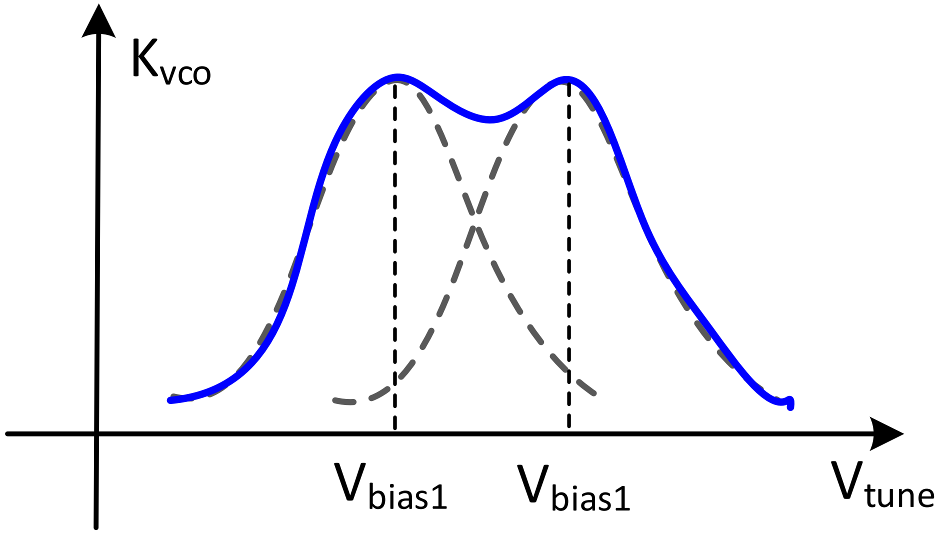
$$K_{VCO} = \frac{d\omega_{osc}}{dV} = \frac{d\omega_{osc}}{dC} \cdot \frac{dC}{dV} = -\frac{1}{2} L \frac{1}{\left(\sqrt{LC}\right)^3} \frac{dC}{dV} = -\frac{L}{2} \cdot \omega_{osc}^3 \frac{dC}{dV}$$

- VCO gain is very nonlinear
  - K_{VCO} over V_{tune} nonlinearity due to varactor C ~ V nonlinear characteristics
  - K_{VCO} over ω_{osc} due to capacitance tuning
- Impact of VCO gain nonlinearity
  - PLL loop gain variation
  - Phase modulation nonlinearity
  - Phase noise degradation at higher ω_{osc} with larger K_{VCO}
- VCO gain linearization over V_{tune}
  - Varactor equalization [Ref 12]
  - Inductive tuning [Ref 13]
  - Feedback scheme [Ref 14]
- VCO gain linearization over ω_{osc}
  - Programmable varactor [Ref 15]



VCO Gain Linearization





- Bias two or more varactors at different bias voltages
  - Shift C ~ V current with different offset
  - Total K_{VCO} is determined by the sum of each varactor K_{VCO}
  - A flat top K_{VCO} instead of a bell shaped K_{VCO}
- Linearization is PVT and mismatch sensitive

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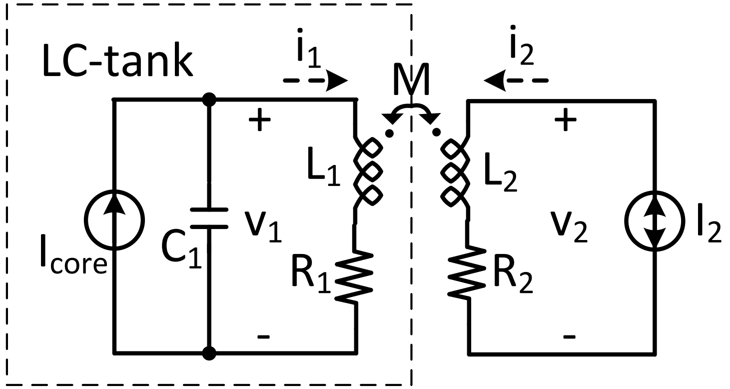
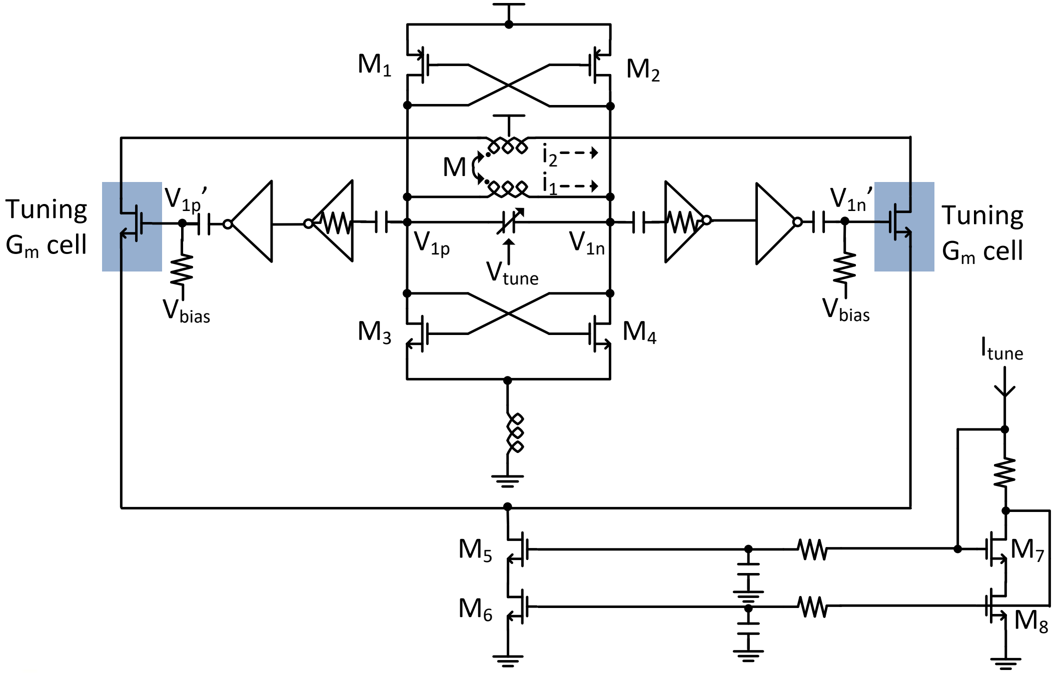
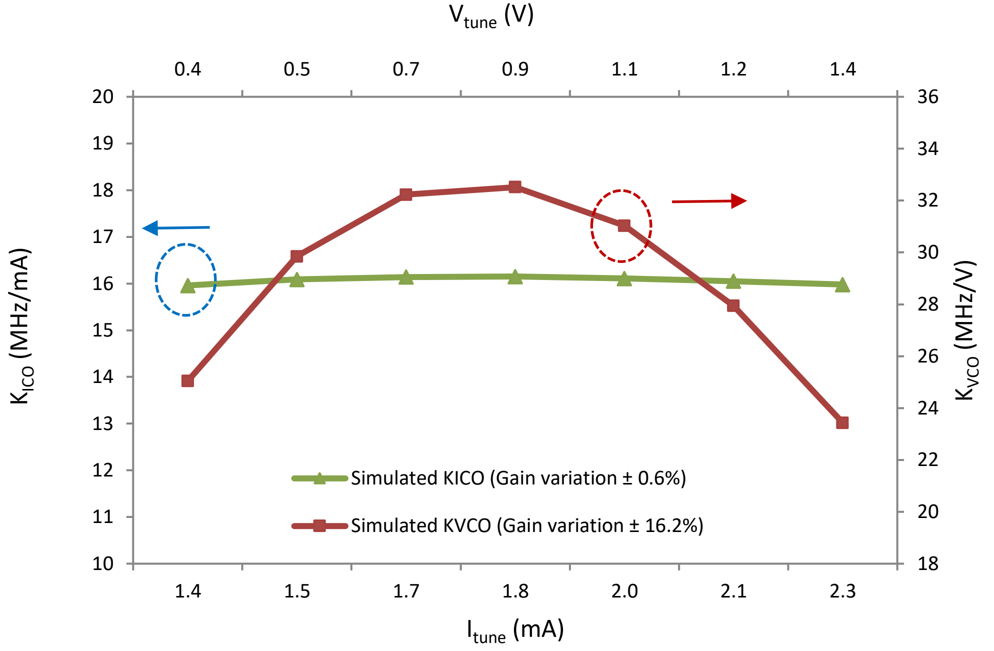
VCO Gain Linearization





- Transformer-based inductive frequency tuning
- $v_1 = s(L_1 + \alpha M)i_1 + R_1 i_1,\ \alpha = \frac{i_2}{i_1} = \frac{i_2}{Q \cdot I_{core}}$
- $f_{osc} = \frac{1}{2\pi\sqrt{(L_1+\alpha M)C_1}} = \frac{1}{2\pi\sqrt{L_1 C_1}} \frac{1}{\sqrt{1+\frac{i_2 M}{Q \cdot I_{core} L_1}}} \approx f_0\left(1 - \frac{1}{2}\frac{i_2 M}{Q \cdot I_{core} L_1}\right)$
- $|K_{ICO}| = \left|\frac{df_{osc}}{di_2}\right| = \frac{1}{2} f_0 \frac{M}{Q \cdot I_{core} L_1}$



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